Problem · Unit 5 · 10 points · 60 min · Standard exam level

P05 — Flight pendulum on an accelerating support: inertia forces, equilibrium and a change of regime

A flight-test pendulum hangs from a rail fixed to the ceiling of an aircraft cabin. The support A slides along the rail with a ***constant acceleration*** $\mathbf{a}_A = a_0\,\mathbf{i}$ *with respect to the inertial frame* $S_0 = \{O;\mathbf{i},\mathbf{j},\mathbf{k}\}$ *(x horizontal, forward; y vertical, upward; uniform gravity $-g\,\mathbf{j}$), starting from rest at $t = 0$. The bob P of mass m is joined to A by a massless, inextensible string of length ℓ that stays taut. The angle θ is measured from the downward vertical, positive towards $+x$. At $t = 0$ the bob hangs at rest with respect to the cabin ($\theta = 0$, $\dot\theta = 0$).*

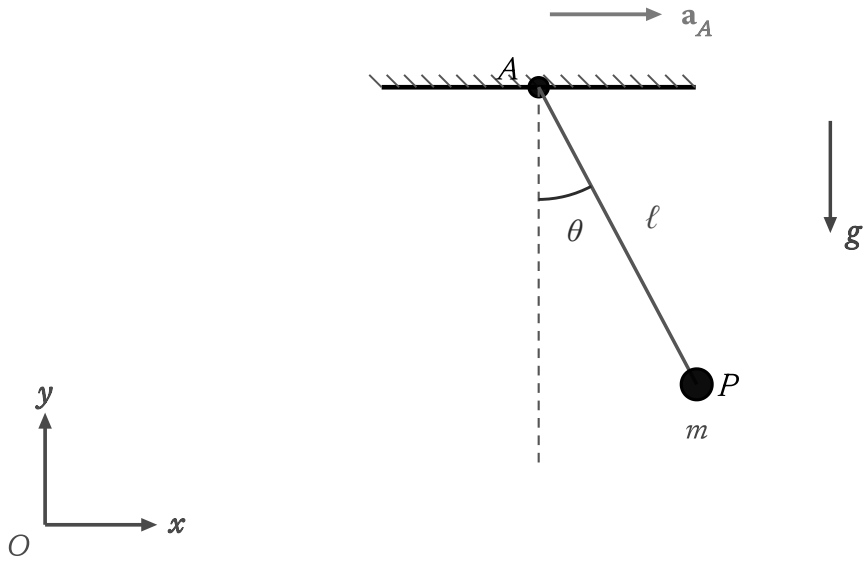


Figure 1. Flight pendulum: support A with acceleration $\mathbf{a}_A = a_0\,\mathbf{i}$, string of length ℓ, bob P of mass m, angle θ from the downward vertical, positive towards $+x$.

Use the reference frame $S_1 = \{A;\mathbf{i},\mathbf{j},\mathbf{k}\}$ that translates with A (its axes stay parallel to those of S_0), and, only to project vectors, the unit vectors $\mathbf{e}_r = \sin\theta\,\mathbf{i} - \cos\theta\,\mathbf{j}$ and $\mathbf{e}_\theta = \cos\theta\,\mathbf{i} + \sin\theta\,\mathbf{j}$.

Setup

- Data: m, ℓ, g, $a_0 > 0$. In the numerical parts (f) and (g): $a_0 = g/\sqrt{3}$.
- Hypotheses: point particle, planar motion, string always taut (checked in (g)), no friction, S_0 inertial.
- S_1 is a **translating** frame: $\boldsymbol{\omega}_{10} = \mathbf{0}$, so the only inertia force is $-m\,\mathbf{a}_A$ (no Coriolis, no centrifugal force).

Part (a) · 1 point

Degrees of freedom and frames

How many effective degrees of freedom has P? State the angular velocity of S_1 with respect to S_0 and the kind of inertia force that appears in S_1.

Step 1. Count

A free particle in the plane has 2 degrees of freedom. The string gives the holonomic constraint $|\mathbf{r}_{AP}| = \ell$, which removes one: θ describes the configuration. The motion of A is given ($\mathbf{a}_A = a_0\mathbf{i}$), so it adds no unknown: the constraint is **rheonomic** in S_0 and scleronomic in S_1.

Step 2. The frame

S_1 keeps its axes parallel to those of S_0, so $\boldsymbol{\omega}_{10} = \mathbf{0}$. In the relative-motion law $\mathbf{F} + \mathbf{F}_{in} = m\mathbf{a}_{rel}$ (Notes §4.9) the Coriolis term $-2m\boldsymbol{\omega} \times \mathbf{v}_{rel}$ and the centrifugal term vanish, and only $\mathbf{F}_{in} = -m\mathbf{a}_A$ remains.

KEY RESULTS

- 1 degree of freedom, coordinate $\theta(t)$.
- $\boldsymbol{\omega}_{10} = \mathbf{0}$; only the translational inertia force $-ma_0\mathbf{i}$.

Part (b) · 1.5 points

Kinematics of P

*Write the position, velocity and acceleration of P **relative to** S_1, and the absolute velocity and acceleration of P in S_0, in the basis $\{\mathbf{e}_r, \mathbf{e}_\theta\}$.*

Change of basis

The projection basis $\{\mathbf{e}_r, \mathbf{e}_\theta\}$ is related to $\{\mathbf{i}, \mathbf{j}\}$ (planar, $\mathbf{k}$ common) by

$$\begin{pmatrix} \mathbf{e}_r \\ \mathbf{e}_\theta \end{pmatrix} = R \begin{pmatrix} \mathbf{i} \\ \mathbf{j} \end{pmatrix}, \qquad R = \begin{pmatrix} \sin\theta & -\cos\theta \\ \cos\theta & \sin\theta \end{pmatrix}$$

$$\det R = \sin^2\theta + \cos^2\theta = 1, \qquad RR^T = \begin{pmatrix} \sin^2\theta + \cos^2\theta & 0 \\ 0 & \cos^2\theta + \sin^2\theta \end{pmatrix} = I$$

so R is orthonormal and right-handed ($\mathbf{e}_r \times \mathbf{e}_\theta = \mathbf{k}$). The inverse is R^T: $\mathbf{i} = \sin\theta\,\mathbf{e}_r + \cos\theta\,\mathbf{e}_\theta$, used below.

Step 1. Derivatives of the unit vectors

From $\mathbf{e}_r = \sin\theta\,\mathbf{i} - \cos\theta\,\mathbf{j}$ and the chain rule $\frac{d}{dt} f(\theta(t)) = \frac{df}{d\theta}\dot\theta$:

$$\dot{\mathbf{e}}_r = \dot\theta\,(\cos\theta\,\mathbf{i} + \sin\theta\,\mathbf{j}) = \dot\theta\,\mathbf{e}_\theta, \qquad \dot{\mathbf{e}}_\theta = \dot\theta\,(-\sin\theta\,\mathbf{i} + \cos\theta\,\mathbf{j}) = -\dot\theta\,\mathbf{e}_r$$

Step 2. Relative motion

$$\mathbf{r}_{rel} = \ell\,\mathbf{e}_r, \qquad \mathbf{v}_{rel} = \ell\dot\theta\,\mathbf{e}_\theta, \qquad \mathbf{a}_{rel} = \ell\ddot\theta\,\mathbf{e}_\theta + \ell\dot\theta\,\dot{\mathbf{e}}_\theta = \ell\ddot\theta\,\mathbf{e}_\theta - \ell\dot\theta^2\,\mathbf{e}_r$$

(ℓ is constant, so the string produces no radial relative velocity.)

Step 3. Absolute motion

The support starts from rest with constant acceleration: $\mathbf{v}_A = a_0 t\,\mathbf{i}$, $\mathbf{r}_A = \frac{1}{2}a_0 t^2\,\mathbf{i}$. Since S_1 does not rotate, $\mathbf{v}_P = \mathbf{v}_A + \mathbf{v}_{rel}$ and $\mathbf{a}_P = \mathbf{a}_A + \mathbf{a}_{rel}$:

$$\mathbf{v}_P = a_0 t\,\mathbf{i} + \ell\dot\theta\,\mathbf{e}_\theta, \qquad \mathbf{a}_P = a_0\,\mathbf{i} + \ell\ddot\theta\,\mathbf{e}_\theta - \ell\dot\theta^2\,\mathbf{e}_r$$

Projecting $\mathbf{i} = \sin\theta\,\mathbf{e}_r + \cos\theta\,\mathbf{e}_\theta$ (the dot products $\mathbf{i}\cdot\mathbf{e}_r = \sin\theta, \mathbf{i}\cdot\mathbf{e}_\theta = \cos\theta$):

$$\mathbf{a}_P = \big(a_0\sin\theta - \ell\dot\theta^2\big)\mathbf{e}_r + \big(a_0\cos\theta + \ell\ddot\theta\big)\mathbf{e}_\theta$$

KEY RESULTS

- $\mathbf{a}_{rel} = \ell\ddot\theta\,\mathbf{e}_\theta - \ell\dot\theta^2\mathbf{e}_r$.
- $\mathbf{a}_P = a_0\mathbf{i} + \mathbf{a}_{rel}$ with $\mathbf{i} = \sin\theta\,\mathbf{e}_r + \cos\theta\,\mathbf{e}_\theta$.

Part (c) · 1.5 points

Forces on P in S_1

Write every force acting on P in S_1 in the basis $\{\mathbf{e}_r, \mathbf{e}_\theta\}$, including the inertia force. Which of them can do work in S_1?

Step 1. Weight

$\mathbf{W} = -mg\,\mathbf{j}$ with $\mathbf{W}\cdot\mathbf{e}_r = -mg(\mathbf{j}\cdot\mathbf{e}_r) = mg\cos\theta$ and $\mathbf{W}\cdot\mathbf{e}_\theta = -mg\sin\theta$:

$$\mathbf{W} = mg\cos\theta\,\mathbf{e}_r - mg\sin\theta\,\mathbf{e}_\theta$$

Step 2. Tension and inertia force

The string is massless and taut, so its force on P points to A: $\mathbf{T} = -T\,\mathbf{e}_r$ with $T \geq 0$ unknown. The inertia force of the translating frame is

$$\mathbf{F}_{in} = -m\,\mathbf{a}_A = -ma_0\,\mathbf{i} = -ma_0\sin\theta\,\mathbf{e}_r - ma_0\cos\theta\,\mathbf{e}_\theta$$

Step 3. Work in S_1

The power of a force is $\mathbf{F}\cdot\mathbf{v}_{rel}$ with $\mathbf{v}_{rel} = \ell\dot\theta\,\mathbf{e}_\theta$: only the $\mathbf{e}_\theta$ components work. $\mathbf{T}\cdot\mathbf{v}_{rel} = 0$. $\mathbf{W}$ and $\mathbf{F}_{in}$ are uniform forces, hence conservative, with potentials $V_g = -mg\ell\cos\theta$ and $V_{in} = ma_0\ell\sin\theta$ (check: $-\partial V/\partial\theta/\ell$ gives their $\mathbf{e}_\theta$ components).

KEY RESULTS

- $\mathbf{W} = mg(\cos\theta\,\mathbf{e}_r - \sin\theta\,\mathbf{e}_\theta)$, $\mathbf{T} = -T\mathbf{e}_r$, $\mathbf{F}_{in} = -ma_0(\sin\theta\,\mathbf{e}_r + \cos\theta\,\mathbf{e}_\theta)$.
- $\mathbf{T}$ does no work (it is orthogonal to $\mathbf{v}_{rel}$); $\mathbf{W}$ and $\mathbf{F}_{in}$ are conservative.

Part (d) · 2 points

Equation of motion and tension

Obtain the equation $\ddot\theta = f(\theta)$ that does not contain T, and the expression of T as a function of θ and $\dot\theta$.

Step 1. Newton in S_1

$$m\big(\ell\ddot\theta\,\mathbf{e}_\theta - \ell\dot\theta^2\mathbf{e}_r\big) = \mathbf{W} + \mathbf{T} + \mathbf{F}_{in}$$

Step 2. Tangential projection

$$m\ell\ddot\theta = -mg\sin\theta - ma_0\cos\theta \implies \ell\ddot\theta = -g\sin\theta - a_0\cos\theta$$

This equation has one unknown, $\theta(t)$, because the tension is orthogonal to $\mathbf{e}_\theta$.

Step 3. Radial projection

$$-m\ell\dot\theta^2 = mg\cos\theta - T - ma_0\sin\theta \implies T = m\big(\ell\dot\theta^2 + g\cos\theta - a_0\sin\theta\big)$$

Step 4. Cross-check with the absolute frame

Newton in S_0 along $\mathbf{e}_r$: $m\,\mathbf{a}_P\cdot\mathbf{e}_r = m(a_0\sin\theta - \ell\dot\theta^2) = \mathbf{W}\cdot\mathbf{e}_r - T = mg\cos\theta - T$, which gives the same T.

KEY RESULTS

- $\ell\ddot\theta = -g\sin\theta - a_0\cos\theta$.
- $T = m\big(\ell\dot\theta^2 + g\cos\theta - a_0\sin\theta\big)$.

Part (e) · 1 point

Equilibrium and small oscillations

Find the equilibrium angle θ_e and the frequency of the small oscillations about it, as functions of a_0, g, ℓ. What does the pendulum indicate?

Step 1. Equilibrium

Setting $\ddot\theta = 0$: $g\sin\theta + a_0\cos\theta = 0 \Rightarrow \tan\theta_e = -a_0/g$. Two solutions differ by π; the stable one hangs below A: $\theta_e = -\arctan(a_0/g)$.

Step 2. Rewrite the equation

With $\cos\alpha = g/g_{eff}$, $\sin\alpha = a_0/g_{eff}$ ($\alpha = \arctan(a_0/g)$) and the identity $\sin(\theta+\alpha) = \sin\theta\cos\alpha + \cos\theta\sin\alpha$:

$$\ell\ddot\theta = -g_{eff}\big(\sin\theta\cos\alpha + \cos\theta\sin\alpha\big) = -g_{eff}\sin(\theta+\alpha)$$

so $\theta_e = -\alpha$. Define $\varphi = \theta - \theta_e = \theta + \alpha$: $\ell\ddot\varphi = -g_{eff}\sin\varphi$, an ordinary pendulum in a gravity g_{eff}.

Step 3. Linearisation

For $|\varphi| \ll 1$, $\sin\varphi \approx \varphi$ and $\ddot\varphi + \dfrac{g_{eff}}{\ell}\varphi = 0$:

$$\omega = \sqrt{\frac{g_{eff}}{\ell}} = \frac{\sqrt[4]{g^2 + a_0^2}}{\sqrt{\ell}}, \qquad \tau = 2\pi\sqrt{\frac{\ell}{g_{eff}}}$$

The pendulum aligns with the **apparent gravity** $\mathbf{g} - \mathbf{a}_A$ (the specific force) and oscillates faster than at rest.

KEY RESULTS

- $\theta_e = -\arctan(a_0/g)$ (the bob lags behind), $g_{eff} = \sqrt{g^2 + a_0^2}$.
- $\omega^2 = g_{eff}/\ell$, stable equilibrium.

Part (f) · 1.5 points

First integral and the swing from $\theta = 0$

Write a first integral of the motion in S_1. For $a_0 = g/\sqrt{3}$, with the initial conditions of the statement, find the extreme angles of the swing and $\ell\dot\theta^2$ when the bob passes θ_e.

Step 1. Energy in S_1

In S_1 the work is done by $\mathbf{W}$ and $\mathbf{F}_{in}$ (conservative, part (c)); $\mathbf{T}$ does none. Hence $E = T_{kin} + V_g + V_{in}$ is constant:

$$\frac{E}{m} = \tfrac{1}{2}\ell^2\dot\theta^2 - g\ell\cos\theta + a_0\ell\sin\theta$$

Check: $\dfrac{d}{dt}\dfrac{E}{m} = \ell\dot\theta\big(\ell\ddot\theta + g\sin\theta + a_0\cos\theta\big) = 0$ by part (d). With $-g\cos\theta + a_0\sin\theta = -g_{eff}\cos(\theta + \alpha)$:

$$\frac{E}{m} = \tfrac{1}{2}\ell^2\dot\theta^2 - g_{eff}\,\ell\cos\varphi$$

Step 2. Numbers

For $a_0 = g/\sqrt{3}$: $\tan\alpha = 1/\sqrt{3}$, so $\alpha = 30^\circ$, $\theta_e = -30^\circ$ and $g_{eff} = g\sqrt{1 + 1/3} = 2g/\sqrt{3}$. At $t = 0$: $\theta = 0 \Rightarrow \varphi_0 = 30^\circ$, $\dot\theta = 0$.

Step 3. Extremes and speed

Equating E at φ_0 and at a generic φ:

$$\ell^2\dot\theta^2 = 2g_{eff}\,\ell\,\big(\cos\varphi - \cos 30^\circ\big)$$

It vanishes at $\varphi = \pm 30^\circ$, i.e. $\theta = 0$ and $\theta = -60^\circ$. At $\varphi = 0$ the factor is $\cos 0 - \cos 30^\circ = 1 - \cos 30^\circ$:

$$\ell\dot\theta^2\Big|_{\theta_e} = 2g_{eff}\,(1-\cos 30^\circ) = \frac{4g}{\sqrt3}\Big(1-\frac{\sqrt3}{2}\Big) = \Big(\frac{4}{\sqrt3}-2\Big)g \approx 0.309\,g$$

KEY RESULTS

- $\frac12\ell^2\dot\theta^2 - g\ell\cos\theta + a_0\ell\sin\theta = \text{const}$, i.e. $\frac12\ell^2\dot\theta^2 - g_{eff}\ell\cos\varphi = \text{const}$.
- $\theta_e = -30^\circ$, extremes $\theta = 0$ and $\theta = -60^\circ$; $\ell\dot\theta^2(\theta_e) = 2g_{eff}(1-\cos 30^\circ) = (4/\sqrt3 - 2)\,g \approx 0.309\,g$.

Part (g) · 1.5 points

Tension and a change of regime

For $a_0 = g/\sqrt3$: (i) compute T at θ_e and show that the string never goes slack. (ii) At the instant the bob passes θ_e for the first time (moving backwards, $\dot\theta < 0$) the support stops accelerating ($a_0 \to 0$, A keeps its velocity). Find the maximum angle of the following motion and the tension when the bob passes the vertical.

Step 1. Tension along the swing

From (d), $T = m(\ell\dot\theta^2 + g\cos\theta - a_0\sin\theta)$ and $g\cos\theta - a_0\sin\theta = g_{eff}\cos\varphi$. With $\ell\dot\theta^2 = 2g_{eff}(\cos\varphi - \cos 30^\circ)$ of (f):

$$T(\varphi) = mg_{eff}\big(3\cos\varphi - 2\cos 30^\circ\big)$$

At $\varphi = 0$: $T = m\frac{2g}{\sqrt3}(3-\sqrt3) = mg(2\sqrt3-2) \approx 1.464\,mg$. T decreases with $|\varphi|$; its minimum is at $\varphi = \pm 30^\circ$: $T = mg_{eff}\cos 30^\circ = mg > 0$. The string is taut during the whole swing.

Step 2. Initial conditions of phase 2

At the switch: $\theta_1 = -30^\circ$, $\ell\dot\theta_1^2 = (4/\sqrt3 - 2)g$. Afterwards $\ell\ddot\theta = -g\sin\theta$ (ordinary pendulum), with the energy per unit mass $\frac12\ell^2\dot\theta^2 - g\ell\cos\theta = \text{const}$. The bob has kinetic energy at θ_1, which is **not** the equilibrium of the new motion.

Step 3. Maximum angle

Setting $\dot\theta = 0$ at θ_m:

$$-g\ell\cos\theta_m = \tfrac12\ell^2\dot\theta_1^2 - g\ell\cos 30^\circ \implies \cos\theta_m = \frac{\sqrt3}{2} - \frac12\Big(\frac{4}{\sqrt3}-2\Big) = 1 - \frac{1}{2\sqrt3} = 1 - \frac{\sqrt3}{6}$$

so $\theta_m = \arccos(0.7113) = 44.66^\circ$: the bob, which was at -30° moving backwards, first reaches -44.66° (beyond the old equilibrium), then swings through the vertical to $+44.66^\circ$: the new motion is symmetric, $\theta \in [-44.66^\circ, +44.66^\circ]$, about the vertical.

Step 4. Tension at the vertical

The energy $\frac{1}{2}\ell^2\dot\theta^2 - g\ell\cos\theta$ is the same at $\theta_1 = -30^\circ$ and at $\theta = 0$, so $\ell\dot\theta^2(0) = \ell\dot\theta_1^2 + 2g(\cos 0 - \cos 30^\circ)$ with $\cos 0 - \cos 30^\circ = 1 - \sqrt{3}/2$:

$$\ell\dot\theta^2(0) = \Big(\frac{4}{\sqrt{3}} - 2\Big)g + 2g\Big(1 - \frac{\sqrt{3}}{2}\Big) = \Big(\frac{4}{\sqrt{3}} - \sqrt{3}\Big)g = \frac{g}{\sqrt{3}}$$

and with $a_0 = 0$: $T(0) = m\big(\ell\dot\theta^2 + g\big) = mg\big(1 + \frac{1}{\sqrt{3}}\big) \approx 1.577\,mg$.

KEY RESULTS

- $T(\varphi) = mg_{eff}(3\cos\varphi - 2\cos 30^\circ)$; $T(\theta_e) = mg(2\sqrt{3} - 2) \approx 1.46\,mg$; $T_{min} = mg$ at the turning points.
- Phase 2: $\cos\theta_{max} = 1 - \sqrt{3}/6$, $\theta_{max} \approx 44.66^\circ$ (the swing is $\pm 44.66^\circ$ about the vertical; the first turning point is **behind** the vertical, at -44.66°); $T(0) = mg(1 + 1/\sqrt{3}) \approx 1.58\,mg$.

Interpretation and checks

- **Dimensional check:** a_0, g and $\ell\dot\theta^2$ are accelerations (m s^{-2}); $\omega^2 = g_{eff}/\ell$ is s^{-2}; T is $m\times$ acceleration = N.
- **Limits:** $a_0 \to 0$ gives $\theta_e = 0$, $g_{eff} = g$ and the ordinary pendulum; $a_0 \gg g$ gives $\theta_e \to -90^\circ$ (the string lies horizontal, behind the support) and $\omega^2 \approx a_0/\ell$.
- **Common mistakes in the exam:** (1) writing the inertia force with the wrong sign (it is $-m\mathbf{a}_A$, opposite to the support's acceleration); (2) forgetting that $\mathbf{T}$ is orthogonal to $\mathbf{v}_{rel}$ and adding its work; (3) taking the old equilibrium as the centre of the new motion after the support stops: the bob keeps its speed and swings **past** the vertical; (4) using g instead of g_{eff} in the period.